BTS–I(R)–11–23–2195 Reg. No. 20523616.

C

B.Tech. Degree I Semester Regular Examination November 2023

CE/ME/SE 23-200-0105 A BASIC MECHANICAL ENGINEERING

(2023 *Scheme*)

Time: 3 Hours Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Summarize the role of mechanical engineering, different energy sources, and basic thermodynamic laws.

CO2: Illustrate the principles and types of power-generating and power producing devices.

CO3: Explain the working of power transmission systems, electric and hybrid vehicles, and modern fuel injection systems.

CO4: Demonstrate the types and classification of composite and smart materials and joining processes.

CO5: Summarize the machine tools operations and advanced manufacturing systems.

CO6: Explain the concepts of Mechatronics, Robotics, and Automation in IoT.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

		$(5 \times 2 = 10)$	Marks	BL	CO	PO
I.	(a)	What are the desirable properties of a refrigerant?	2	L2	2	1
	(b)	Compare the merits and demerits of Electric and Hybrid vehicles.	2	L2	3	1
	(c)	What is Additive Manufacturing? List the steps in Additive manufacturing.	2	L2	4	1
	(d)	Differentiate between open-loop and closed-loop mechatronic systems.	2	L2	6	1
	(e)	"No engine can be made to work on Carnot cycle", Comment on this.	2	L2	1	1,2

PART B

		$(4 \times 10 = 40)$				
II.	(a)	Explain the structure and working of a 4- stroke diesel engine with adequate sketches.	7	L2	2	1,2,3
	(b)	"A process must satisfy 1st law of thermodynamics, but satisfying 1st law alone does not ensure that the process will actually takes place". Do you agree with the above statement? If yes, explain why and how it is resolved with the help of an example.	3	L2	1	1,2,3
		OR				
III.	(a)	Explain the working of house hold refrigerator with neat sketches. Explain working with vapour compression refrigeration system.	7	L3	2	1,2,3
	(b)	Compare fire tube and water tube boilers.	3	L2	2	1,2,3

(P.T.O.)

Q. No.		Question	Marks	BL	CO	PO
IV.	(a)	With the help of block diagrams explain CRDi and MPFI fuel systems.	6	L2	3	1
	(b)	The power transmitted by an open belt drive with the following specifications is 1570 W. Tight side tension = 700 N Diameter of Driver pulley = 0.5 m Speed of Driver pulley = 200 rpm (i) Find the slack side tension. (ii) Find the speed of driven pulley with 100 mm diameter.	4	L3	3	1,2,3
		OR				
V.	(a)	What is the need of cooling system in automobiles? Explain the types of cooling system and their application with the help of neat sketches.	6	L2	3	1,2,3
	(b)	Explain any two lubrication systems in an IC engine with neat sketches.	4	L2	3	1,2,3
VI.	(a)	Explain Arc welding with adequate figures.	6	L2	4	1,2,3
	(b)	Explain Up-milling and Down-milling with neat sketches.	4	L2	5	1,2,3
		OR				
VII.	(a)	Explain the components of CNC with neat figure.	6	L2	5	1,2,3
	(b)	Explain Gas welding and the types of flames in gas welding operation with figures.	4	L2	4	1,2,3
VIII.	(a)	Discuss the basic elements of automation in industry with block diagrams. Explain the types of automation.	6	L2	6	1,2,3
	(b)	Explain the importance and working of IOT with its practical application.	4	L2	6	1,2,3
		OR				
IX.	(a)	Explain with neat sketches the classification of robotics based on its configuration. Also discuss the work volumes of each type.	10	L2	6	1,2,3

Bloom's Taxonomy Levels
L2 – 87.8%, L3 – 12.2%.

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